

Data Validation Report

Project: M5 Sales Forecasting & Machine Learning Preparation

Date of Report: June 23, 2026

1. Dataset Overview

- The dataset has been successfully prepared and validated for time-series forecasting (Prophet) and machine learning modeling.
- Dataset Name: M5 Sales Forecasting Dataset
- Purpose: Data preparation, feature engineering, and predictive modeling.
- Date Range: January 29, 2011 to April 24, 2016

2. Integrity & Quality Checks

- Missing Values Check
 - Initial Status: Missing values were detected during the initial profile.
 - Resolution: Missing values were handled systematically by replacing them with "None" or appropriate placeholders where required.
 - Final Status: 0 missing values remaining.
- Duplicate Records Check
 - Initial Status: Dataset scanned for duplicate rows.
 - Total Duplicates Found: 0
 - Final Status: No duplicate rows present in the final dataset.

3. Data Type Validation

Core variables required for time-series and machine learning pipelines were validated for correct schemas:

Column name	Expected Type	Status	Notes
Ds	Date / Datetime	Validated	Primary time index for Prophet
Y	Numeric (Float/Int)	Validated	Target variable (Total Sales)
Engineered algorithm	Mixed (Numeric/Categorical)	Validated	Correctly cast for ML

4. Feature Schema Verification

The following 30 features have been verified and are fully present in the final prepared dataset:

Time & Calendar Features

d (Day identifier)

Month / year / day_of_year / Week_of_year

Day_Of_Week / Weekend

month sin / month cos / day sin / day cos (Cyclic encodings)

Event & Policy Indicators

event_name_1 / event_type_1 / is holiday

snap CA / snap TX / snap WI

wm_yr_wk / snap active / is_snap_day

Demand & Pricing Signals

Target: total sales

Lags: lag_7 / lag_14 / lag_28

Rolling Windows: rolling_mean_7 / rolling_mean_14 / rolling_mean_28

Financials: avg_price / price lag / price change

5. Summary & Next Steps

Validation Status: **PASSED**

The dataset meets all data quality requirements. Structural integrity is intact, features are properly engineered, and schemas are aligned with model inputs.

Data Cleaning: Complete (No duplicates, missing values resolved).

Feature Engineering: Complete (Temporal, cyclical, holiday, and lag features generated).

Readiness: The final dataset is fully optimized and approved for Prophet and Machine Learning forecasting tasks.

Feature Engineering & Modeling Dataset (Report Write-up)

1. Data Preparation Stage

The raw dataset was first cleaned and validated to ensure consistency and reliability for time series forecasting. The dataset was sorted in chronological order based on the date column, and missing or duplicate values were handled appropriately.

2. Feature Engineering Process

To improve the predictive performance of machine learning models, multiple features were created from the original time series data.

- Temporal Features

The following time-based features were extracted from the date column:

Year

Month

Week Number

Day of Week

Weekend Indicator (0/1)

These features help capture seasonal and calendar-based patterns in sales data.

- Lag Features

Lag features were created to capture dependency on historical values:

lag_7: Sales value from 7 days prior

lag_14: Sales value from 14 days prior

lag_28: Sales value from 28 days prior

These features help the model understand short-term and long-term trends in the data.

- Rolling Statistics

A rolling mean feature was created:

rolling_mean_7: Average sales over the past 7 days

This feature helps smooth fluctuations and highlights overall trends in sales behavior.

3. Modeling Dataset Preparation

Two separate datasets were prepared for forecasting:

- Prophet Dataset

A simplified dataset containing:

ds (Date)

y (Total Sales)

This dataset is used for statistical time series forecasting and trend analysis.

- LightGBM Dataset

A feature-rich dataset containing:

Temporal features (Year, Month, Week, etc.)

Lag features (lag_7, lag_14, lag_28)

Rolling statistics (rolling_mean_7)

Target variable (Total Sales)

This dataset is used for machine learning-based forecasting using LightGBM.

4. Outcome

The final datasets are fully prepared and validated for model training. The feature engineering process significantly enhances the dataset by adding temporal, lag-based, and statistical insights, enabling both Prophet and LightGBM models to generate accurate sales forecasts.